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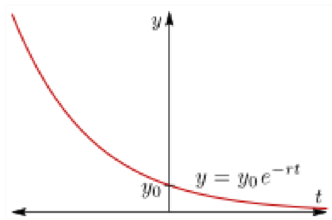
22MA201/ COMMON TO ALL BRANCHES (EXCEPT CSBS)

UNIT I – FIRST ORDER LINEAR DIFFERENTIAL EQUATIONS

LP 07 – EXPONENTIAL DECAY

THE EXPONENTIAL DECAY EQUATION:

The exponential growth equation is the differential equation $\frac{dy}{dx} = -ky, (k < 0)$. Its solutions are exponential functions of the form $y = Ce^{-kt}$, where $y_0 = y(0) = C$ is the initial value of y. Here 'k' is the decay constant.



▲ **Figure 5:** Exponential decay.

Note:

Half life is the time required for a quantity to reduce to half its initial value. The term is commonly used in a nuclear physics to describe how quickly unstable atoms undergoes, or how long stable atoms survives, radioactive decay.

The half-life of y is the amount of time that it takes for the value of y to be cut in half. It can be found by solving the equation $e^{-kt} = 1/2$ for t.

Problems :

1. A certain radioactive material is known to decay at a rate proportional to the amount present. If initially there is 50 milligrams of the material present and after two hours it is observed that the material has lost 10% of its original mass, find (a) an expression for the mass of the material at any time 't', (b) the mass of the material after four hours, and (c) the time at which the material has decayed to one half of its initial mass.

Solution:

Let N denote the amount of material present at time 't'. Then

$$\frac{dN}{dt} - kN = 0 .$$

This differential equation is separable and linear , its solution is $N(t) = Ce^{kt}$ ------→(1)

At t=0, we are given that N = 50. Therefore, from (1),

$$50 = Ce^{k(0)} \Rightarrow C = 50.$$

Thus, $N(t) = 50e^{kt}$ --------------→(2)

At t =2, 10% of the original mass of 50mg or 5 mg, has decayed .

Hence, at t = 2, N = 50-5 = 45. Substituting these values in (2) and solving for 'k', we have

$$45 = 50e^{2k}$$
$$\Rightarrow e^{2k} = \frac{45}{50}$$

Taking ln on both sides, we get $k = \frac{\ln 0.9}{2} = -0.053$

Substituting this value into (2), we obtain the amount of mass present at any time t as

$N(t) = 50e^{-0.053t}$ ---------------------→(3)

Where 't' is measured in hours.

We require N at t =4. Substituting t =4 in eq.(3), we get,

$$N(t) = 50e^{-0.053(4)} = 50(0.809) = 40.5 \text{ mg}$$

We require t when N =50/2 =25. Sub. N=25 in eq.(3) and solving for t, we get

$$25 = 50e^{-0.053t} \Rightarrow \frac{25}{50} = e^{-0.053t}$$

Taking ln on both sides, we get $t = \frac{\ln 0.5}{-0.053} = 13.08 \approx 13$ hours

The time required to reduce a decaying material to one half its original mass is called the half-life of the material. Here the half-life is 13 hours.

Practice Problems:

1. A radioactive isotope has an initial mass 200mg , which two years later is 50mg . Find the expression for the amount of the isotope remaining at any time. What is its half-life? (half-life means the time taken for the radioactivity of a specified isotope to fall to half its original value).
2. The engine of a motor boat moving at 10 m/s is shut off. Given that the retardation at any subsequent time (after shutting off the engine) equal to the velocity at that time. Find the velocity after 2 seconds of switching off the engine

Discussion Questions

Try to answer the following questions

1. Define exponential decay.
2. What is half-life in exponential decay?
3. Give a real time example of exponential decay.
4. The decay constant for the radioactive element strontium 90 is $\lambda = 0.0244$ where time is measured in years. How long will it take for a quantity P_0 of strontium 90 to decay to on-half of its original mass?
5. Ten grams of a radioactive substance with decay constant 0.04 is stored in a vault. Assume that time is measured in days, and let P(t) be the amount remaining at time t.
   (a) Give the formula for P(t)
   (b) Give the differential equation satisfied by P(t).
   (c) How much will remain after 5 days?
   (d) What is the half-life of this radioactive substance?
6. The decay constant for the radioactive element cesium 137 is 0.023 when time is measured in years. Find its half-life.